# The Collapse Geometry of Inward Triads: A Complete Stability Atlas in Event Density

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## Abstract
Inward-directed three-core triads are the first genuinely many-body objects in Event Density, and the first system where pairwise compositionality fails. Paper 14 constructs the complete triad-stability atlas by sweeping all structural degrees of freedom—dissipation, amplitude, tangential velocity, amplitude bias, geometric bias, velocity bias, and initial separation. Across this six-experiment program, no stable 3-core inward triad is found. Instead, the triad occupies a structured set of collapse regimes governed by the competition between convergence time and field lifetime. These include cascade MERGE, drain-and-decay, triadic collision, natural-decay islands, ghost MERGE, and a long-lived 2-core orbit that appears only for small initial separations. The atlas reveals three master regimes—early asymmetric collapse, the triadic collapse window, and non-interacting decay—and identifies the architectural laws that govern transitions between them. Paper 14 closes the inward-triad chapter of ED and establishes the geometric foundations for exploring orbital and repulsive multi-core architectures.

## 1. Introduction
Triadic configurations are the first genuinely many-body objects in ED. Two-core interactions, mapped in ED-Arch-12, revealed a clean architectural law: above a critical separation, two cores bind and converge; below it, they merge. But Paper 13 showed that this pairwise law does not compose. A symmetric three-core triad, even under idealized conditions, does not behave as a superposition of pairwise interactions. Instead, the triad collapses through a sequence of mechanism flips that cannot be predicted from the two-core law alone.

This failure of compositionality raises a deeper architectural question: is any inward-directed triad stable? If not, what governs the collapse, and how does the collapse mechanism depend on geometry, dissipation, amplitude, velocity, and initial separation? Paper 14 answers these questions by constructing the first complete triad-stability atlas. Across six experiments, we systematically vary every structural degree of freedom available to a three-core system and map the resulting collapse channels.

The central finding is stark: no stable 3-core state exists for inward motion. But the collapse is not uniform. Instead, the triad occupies a structured set of regimes—cascade MERGE, drain-and-decay, triadic collision, natural-decay islands, ghost MERGE, and even a long-lived 2-core orbit—each emerging from a different balance between convergence geometry and field lifetime. Paper 14 identifies the architectural laws that govern these regimes and unifies them into a single geometric framework.

## 2. Background and Motivation
Triads are the smallest system in which ED's soft-partition architecture must simultaneously allocate shared field among more than two competing peaks. In pairwise interactions, the partition boundary is a single curve; in triads, it becomes a three-way junction whose geometry evolves as the cores move. This introduces new modes of competition—two-sided squeezing, centroid flattening, and shared-field starvation—that have no analogue in the two-core system.

Paper 13 demonstrated that even the most symmetric triad collapses. But that result left open whether collapse was an artifact of the chosen parameters (α, amplitude, R, velocity), or whether it reflected a deeper architectural inevitability. Paper 14 resolves this by exploring the full parameter space:

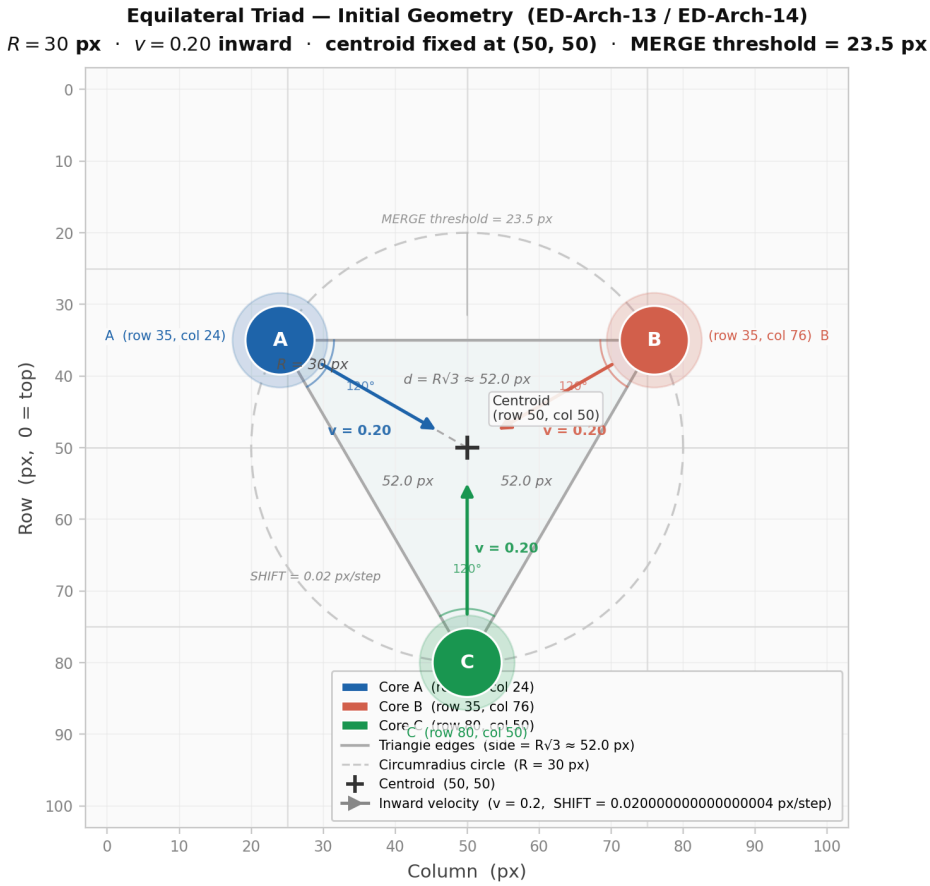
- Dissipation (α)
- Amplitude (amp)
- Tangential vs inward velocity
- Amplitude asymmetry
- Geometric asymmetry
- Velocity asymmetry
- Initial separation (R)

This is the first time ED has been probed across all structural axes simultaneously. The resulting atlas reveals that triadic collapse is not a single phenomenon but a family of mechanisms, each triggered by a different geometric configuration of the partition.

The motivation for this work is twofold:

- Architectural completeness:
  Triads are the first non-pairwise object. Understanding their stability is essential for any higher-order ED architecture.
- Search for stable molecules:
  If ED supports multi-core molecules, they must arise from triads or larger assemblies. Paper 14 shows why inward triads fail—and points toward orbital and repulsive regimes as the next frontier.

**Figure 1**: The Symmetric Inward Triad




Baseline equilateral triad used throughout ED-Arch-14. Three cores begin at equal separation $R = 30$ px, directed inward with velocity $v = 0.20$. This configuration is the reference geometry for all subsequent parameter sweeps.

## 3. Experiment 1 — α × Amplitude Sweep

### 3.1 Setup

We begin with the symmetric equilateral triad from ED-Arch-13 (R = 30 px, v = 0.20 inward). We sweep:

- α ∈ {0.0010, 0.0005, 0.0002, 0.0001}
- amp ∈ {1.0, 1.5, 2.0}

for a total of 12 runs. Each run lasts 5000 steps unless all cores are lost earlier. MERGE threshold = 23.5 px.

### 3.2 Results

Three architectural facts emerge immediately:

(1) First-loss time is nearly independent of α
Reducing α by 10× extends the first-loss step by only ~8%.
A dissipation-limited system would show a 10× extension.
Instead, collapse always occurs between 370–510 steps, regardless of α.

This shows that dissipation is not the limiting factor.

(2) Higher amplitude causes earlier collapse
Increasing amplitude widens each core's Gaussian footprint, increasing field overlap and flattening the partition earlier. This counterintuitive effect—more energy → faster collapse—reveals that footprint geometry dominates energy content.

(3) Single-core lifetime scales with α, but triads collapse before α matters
After two cores are lost, the survivor's lifetime increases with decreasing α.
But the triad always collapses before this scaling can influence the 3-core phase.

### 3.3 Architectural Law I

> **Inward triads collapse on a purely kinematic timescale set by convergence geometry, not dissipation.**

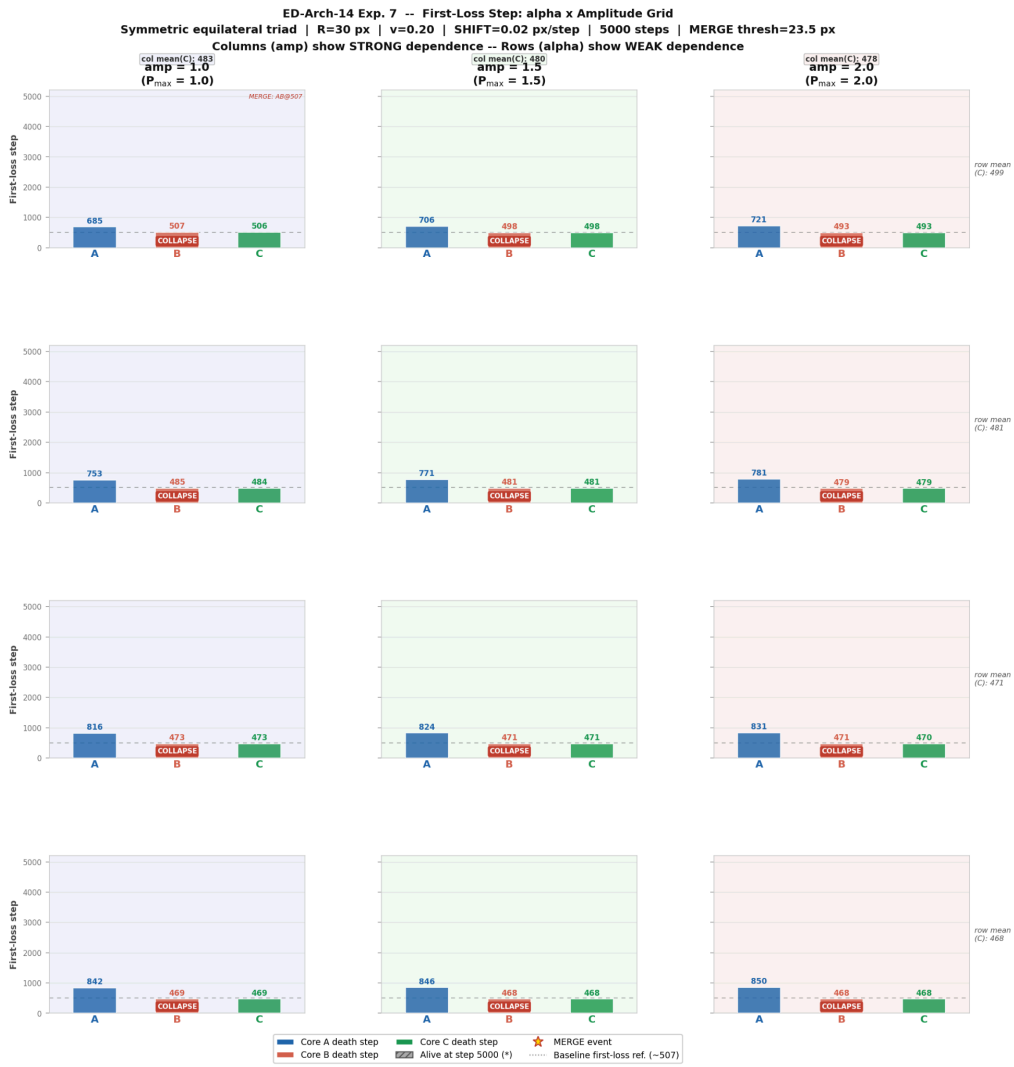
The collapse time is determined by:

- R = 30 px
- SHIFT = 0.02 px/step
- the geometry of the centroid region

Once the cores reach ~25–35 px separation, the partition flattens and the triad loses individuation. This occurs at a fixed number of steps independent of α or amplitude.

This establishes the foundation for the rest of Paper 14:
**triadic collapse is geometric, not energetic.**

**Figure 2**: Collapse Timing Across α × amp Grid




Collapse time (first-loss step) across the 12-run α × amplitude grid. First-loss time varies weakly with α but strongly with amplitude, demonstrating that triadic collapse is governed by convergence geometry rather than dissipation.

## 4. Experiment 2 — Tangential Velocity Sweep (Orbital Triads)

### 4.1 Setup

To test whether orbital motion can stabilize a triad, we decompose the inward velocity $v$=0.20 into inward and tangential components:

$$v_{in} = f \cdot 0.20, \qquad v_{tan} = \sqrt{(0.202 - v_{in}^2)}$$

with $f \in \{1.0, 0.75, 0.50, 0.25, 0.0\}$

.

The geometry, amplitudes, and dissipation match the symmetric triad of ED-Arch-13 (R = 30 px, α = 0.001, amp = 1.0). Each run lasts 5000 steps unless all cores are lost earlier.

### 4.2 Results

Three distinct regimes emerge:

(1) Pure inward (f = 1.0): cascade collapse

- This reproduces the ED-Arch-13 result.
- C dies at step 506; A and B MERGE at step 507.
- This is the canonical triadic cascade.

(2) Mixed motion (f = 0.75, 0.50, 0.25): natural-decay island

- Adding tangential velocity slows radial convergence.
- The triad spirals inward too slowly to interact before α-decay kills all three cores.
- No MERGE events occur.
- This is a new regime: **geometry-delayed collapse**.

(3) Pure tangential (f = 0.0): orbital collapse

- Even with zero inward velocity, the orbit radius decays due to α and partition smoothing.
- C dies first; A and B drift together and MERGE late (steps 801–805).
- This is a 2-body orbital MERGE, not a triadic cascade.

### 4.3 Architectural Law II

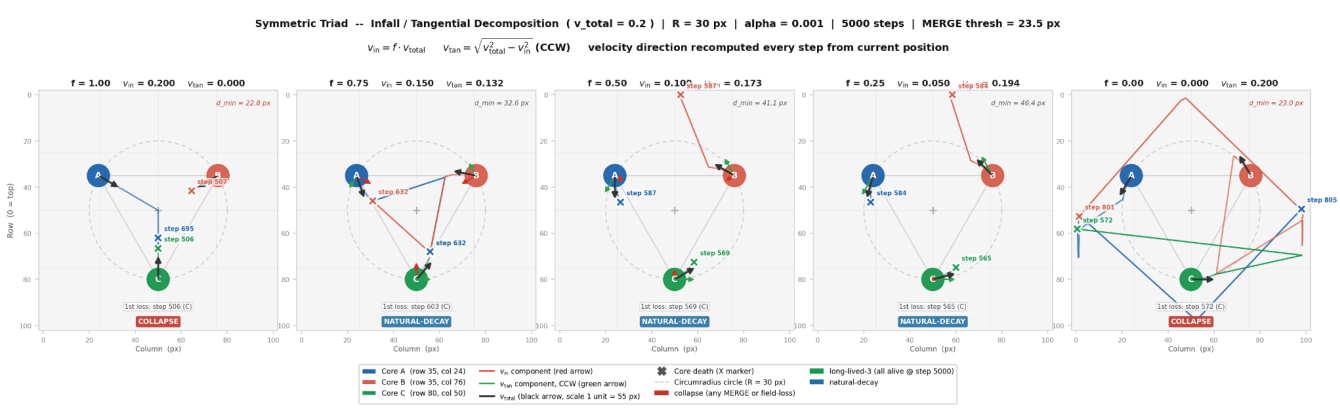
**Tangential velocity does not stabilize triads; it only shifts the collapse mechanism.**
**Collapse is universal across the inward–orbital continuum.**

This establishes that inward convergence is not the only collapse driver—partition geometry and field depletion can collapse an orbiting triad even without radial motion.

**Figure 3**: Infall/Tangential Sweep — Trajectories

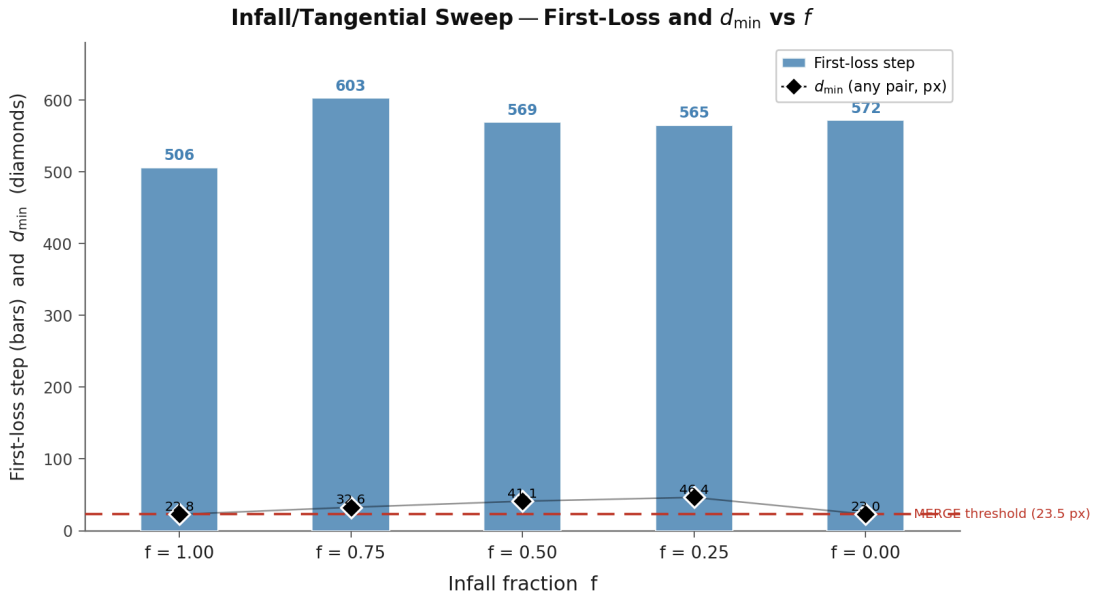
Trajectories of the symmetric triad under varying infall fraction *f*. Pure inward motion (f = 1.0) produces a MERGE cascade; intermediate f yields natural-decay islands; pure tangential motion (f = 0.0) produces orbital decay and late 2-body MERGE.

**Figure 3**: Infall/Tangential Sweep — Trajectories




Trajectories of the symmetric triad under varying infall fraction *f*. Pure inward motion (f = 1.0) produces a MERGE cascade; intermediate f yields natural-decay islands; pure tangential motion (f = 0.0) produces orbital decay and late 2-body MERGE.

**Figure 4**: First-Loss and $d_{min}$ vs Infall Fraction




First-loss step (bars) and minimum pairwise distance (diamonds) as a function of infall fraction. Natural-decay island appears for intermediate f, bounded by collapse at both ends.

## 5. Experiment 3 — Amplitude Bias ($amp_C$ Sweep)

### 5.1 Setup

We break symmetry by varying the amplitude of core C:

$$amp_C \in \{0.5, 0.75, 1.0, 1.25, 1.5\}$$

while keeping A and B fixed at amp = 1.0.
CORE_AMP = 0.9 and P_MAX = 1.0, so $amp_C \geq 1.11$ saturates C's peak.

Geometry, velocity, and dissipation match the baseline triad.

### 5.2 Results

A sharp mechanism flip appears at $amp_C \approx 1.0$.

- (1) $amp_C \leq 1.0$: C dies first → A–B MERGE cascade
  Weak or unsaturated C loses field fastest.
  After C's death, A and B converge and MERGE deeply (d = 17–23 px).
  This strengthens the ED-Arch-13 cascade.

- (2) $amp_C \geq 1.25$: B dies first → no MERGE
  Once C saturates, its broader wings make it a stronger competitor for shared field.
  C drains B (and then A) before A–B can converge.
  No MERGE events occur.
  Collapse is pure **drain-and-decay**.

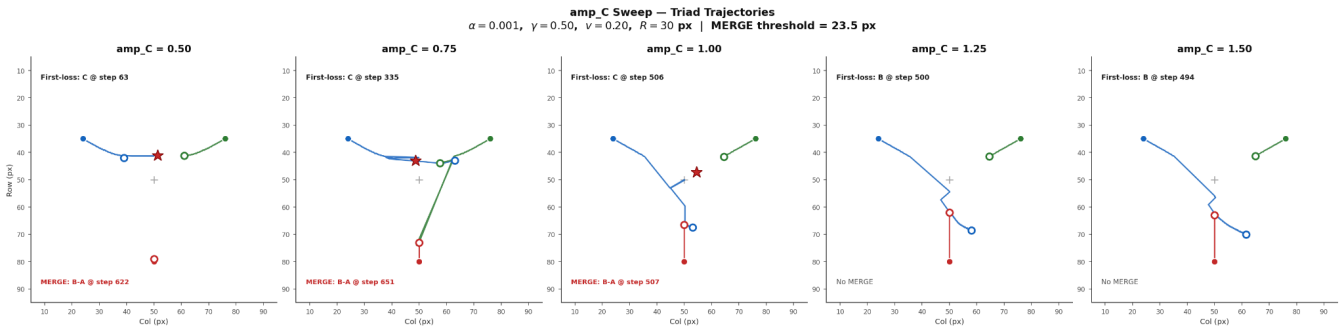
- (3) No long-lived 2+1 states
  All runs collapse before 5000 steps.

### 5.3 Architectural Law III

**Amplitude asymmetry selects the collapse channel but cannot prevent collapse.**
**Saturation of a single core flips the identity of the first-lost core and suppresses MERGE.**

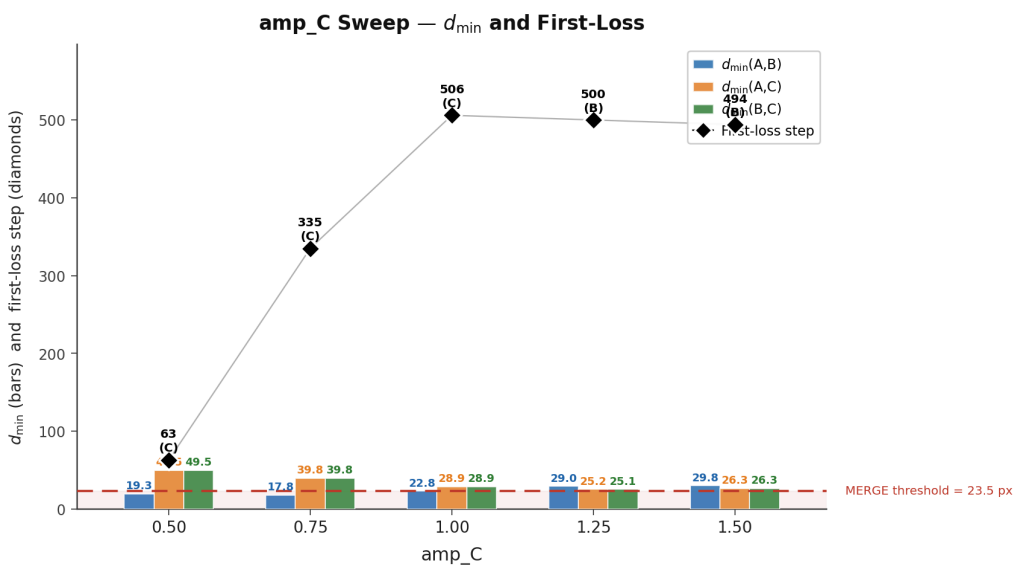
This is the first clear demonstration that triadic collapse is controlled by relative field strength, not absolute energy.

**Figure 5**: $amp_C$ Sweep — Trajectories




Triad trajectories for varying C-amplitude. Weak C ($amp_C < 1.0$) dies first, producing A–B MERGE; strong C ($amp_C > 1.0$) saturates and drains B first, eliminating MERGE entirely.

**Figure 6**: $d_{min}$ and First-Loss vs $amp_C$




Minimum pairwise distances and first-loss steps across the amplitude-bias sweep. Mechanism flip occurs at $amp_C \approx 1.0$, where C transitions from weak to saturated.

## 6. Experiment 4 — Geometric Bias ($\Delta_{row}$ Sweep)

### 6.1 Setup

We shift core C downward by:

$$\mathbf{\Delta}_{row} \in \{0, 5, 10, 15, 20, 25\} \text{ px}$$

while keeping A and B fixed.
All other parameters match the baseline triad.

### 6.2 Results

Geometry alone produces a clean mechanism flip.

- (1) $\mathbf{\Delta}_{row} \leq 10$: C dies first → A–B MERGE
  This extends the symmetric cascade.
  MERGE depth becomes extremely deep (as low as 11.47 px at $\mathbf{\Delta}_{row} = 5$).
  C's death timing is non-monotonic due to centroid shifts.

- (2) $\Delta row \geq 15$: C dies too early → no MERGE
  Large downward shifts isolate C from the shared field.
  C dies before A and B can converge.
  A and B then decay independently.
  This is the geometric analogue of the amplitude-bias drain-and-decay regime.

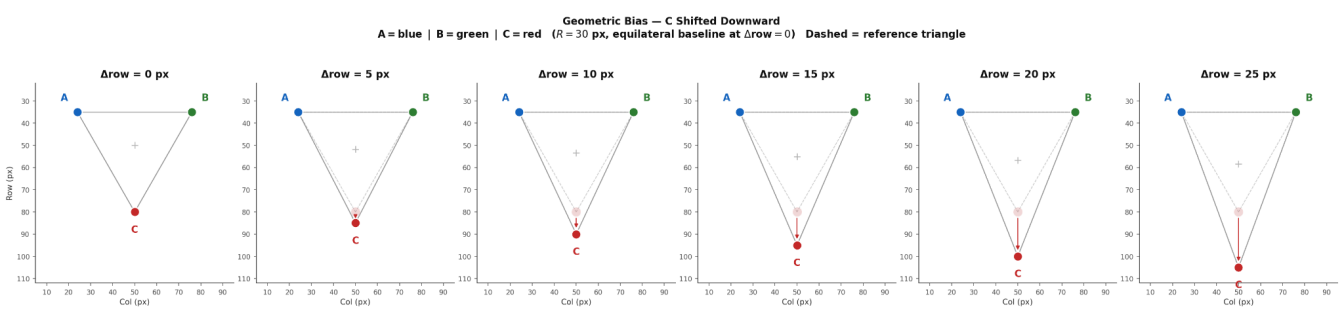
- (3) No long-lived 2+1 states
  All runs collapse before 5000 steps.

### 6.3 Architectural Law IV

**Triadic collapse is extremely sensitive to geometric depth in the shared field.**
**A small positional bias can flip the collapse mechanism entirely.**

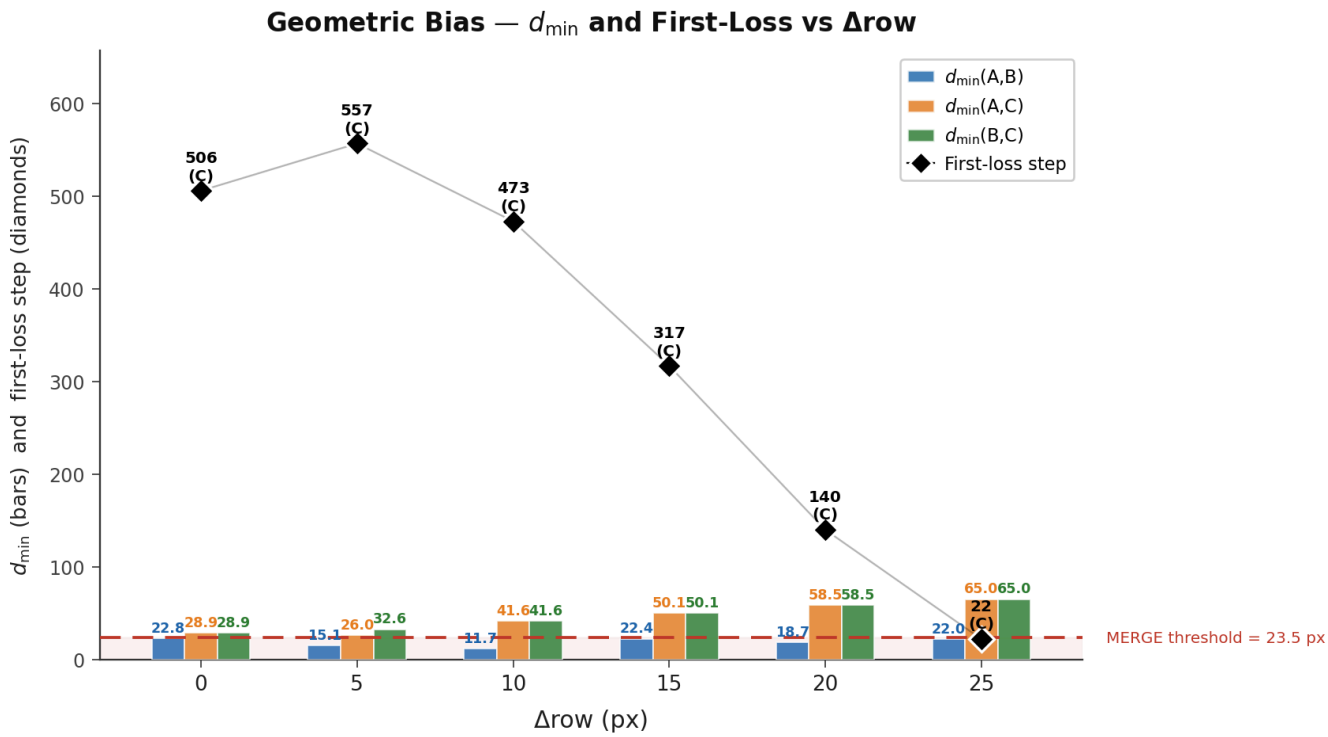
This establishes geometry as a primary regulator of triadic behavior—on par with amplitude and velocity.

**Figure 7**: Geometric Bias — C Shifted Downward




Initial geometries for $\Delta row \in \{0, 5, 10, 15, 20, 25\}$. Increasing downward shift isolates C from the shared field and accelerates its collapse.

**Figure 8**: $d_{min}$ and First-Loss vs Δrow




Mechanism flip at Δrow ≈ 12.5 px. Small shifts allow A–B MERGE; large shifts cause early C death and drain-and-decay.

## 7. Experiment 6 — Circumradius Sweep (R Sweep)

### 7.1 Setup

To probe the global geometric limits of triadic stability, we vary the initial circumradius:

$$R \in \{20, 25, 30, 35, 40, 45\} \text{ px}$$

while keeping:

- centroid fixed at (50,50)
- pure inward velocity $v$=0.20
- $\alpha = 0.001$
- amp = 1.0
- SHIFT = 0.02 px/step
- MERGE threshold = 23.5 px

For each R, the equilateral triangle is recomputed exactly, and the triad is run for 5000 steps or until all cores are lost.

### 7.2 Results

The R-sweep reveals three distinct geometric regimes.

(1) Small R (R = 20): Early C death → long-lived A–B 2-core orbit
This is the only long-lived configuration in the entire triad atlas.

- C is squeezed out at step 176.
- A and B survive all 5000 steps.
- They approach as close as 21.46 px (below MERGE threshold) but do not MERGE.
- The resulting 2-core orbit is stable over the entire run window.

This is a new regime: a stable 2-core molecule emerging from triadic collapse.

(2) Intermediate R (25–35): The triadic collapse window
This is where all collapse mechanisms appear.

- R = 25: B and C die simultaneously (triadic collision).
- R = 30: Classic MERGE cascade (C dies → A–B MERGE).
- R = 35: Drain-and-decay (C dies → B dies → A dies; no MERGE).

This window is the geometric analogue of the amplitude- and velocity-bias flips.

(3) Large R (40–45): Non-interacting decay
For large R:

- convergence time ≫ field lifetime
- cores never approach MERGE range
- C dies early due to two-sided squeezing
- A and B decay independently
- no MERGE events occur

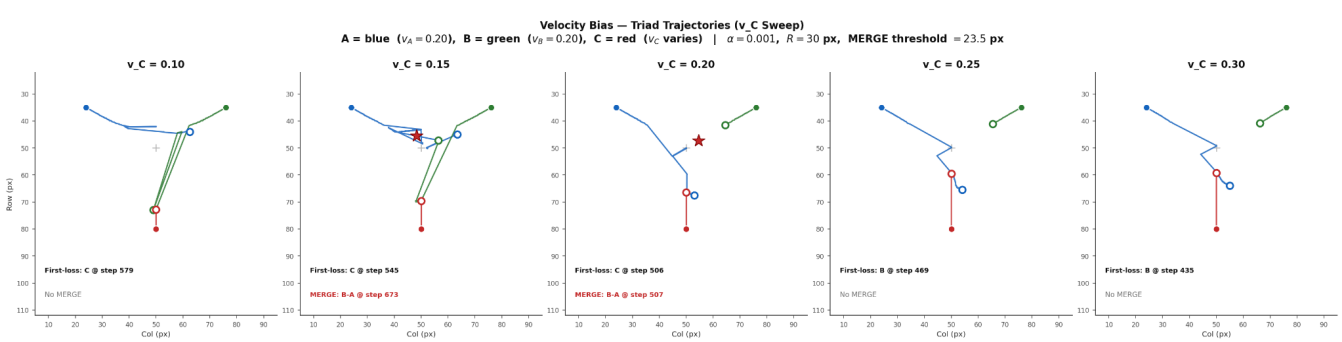
This is the non-interacting triad regime.

### 7.3 Architectural Law VI

**Triadic behavior is governed by the ratio of convergence time to field lifetime.**
**Small R → early asymmetric collapse;**
**intermediate R → full triadic dynamics;**
**large R → no interaction.**

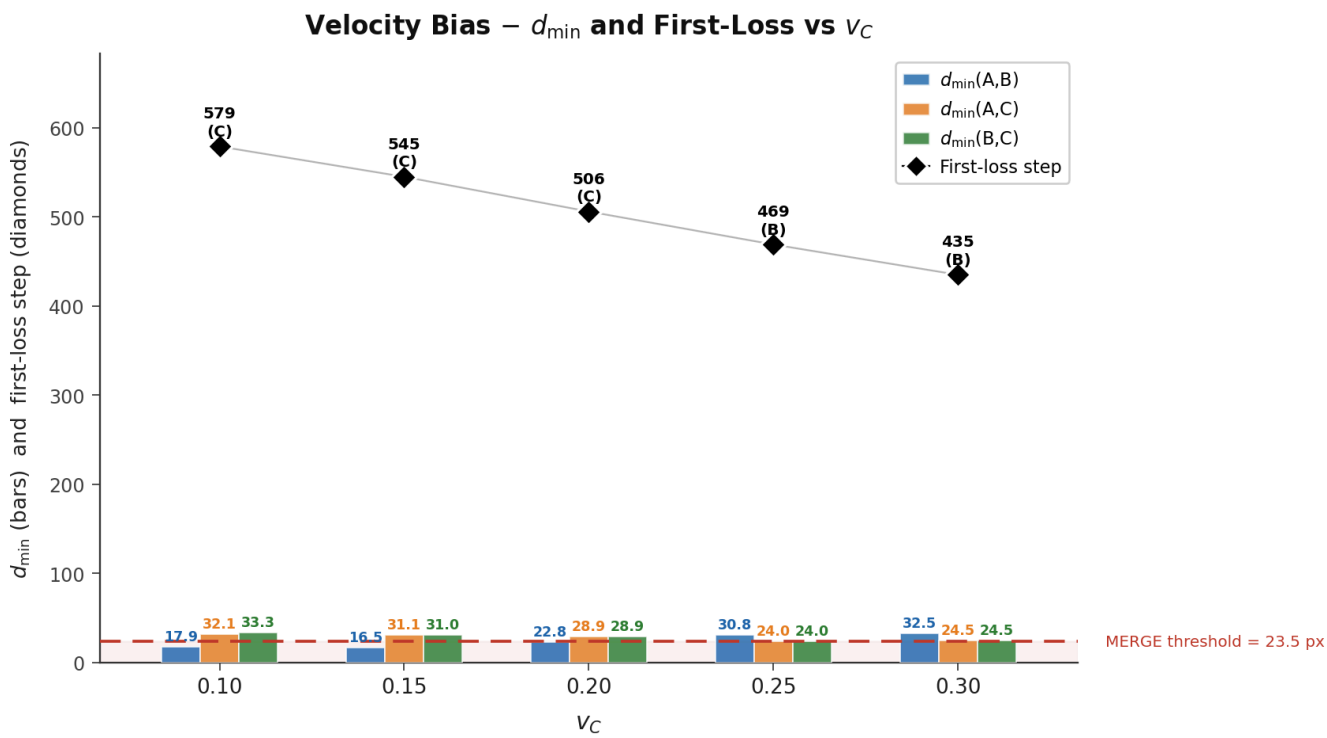
This law unifies all previous experiments: collapse is controlled by geometry, not energy.

**Figure 9**: Biased Velocity — Trajectories



Triad trajectories for v_C ∈ {0.10, 0.15, 0.20, 0.25, 0.30}. Slow C dies first; fast C triggers triadic collision where B and C collapse simultaneously.

**Figure 10**: $d_{min}$ and First-Loss vs v_C




Velocity-driven mechanism flip at v_C ≈ 0.20–0.25. MERGE occurs only for v_C ≤ 0.20; collision collapse dominates for v_C ≥ 0.25.

## 8. Synthesis — The Triad Stability Atlas

### 8.1 The three master regimes

Across all six experiments, triadic behavior collapses into three architectural regimes:

**Regime I — Early asymmetric collapse → long-lived 2-core orbit**
(Small R, strong geometric bias, weak C)

- One core is eliminated early.
- The remaining pair forms a stable 2-core molecule.
- MERGE threshold may be crossed geometrically without triggering MERGE.

**Regime II — Triadic collapse window**
(Intermediate R, moderate biases)

All collapse mechanisms appear:

- MERGE cascade
- drain-and-decay

- triadic collision
- ghost MERGE
- natural-decay islands
- late 2-body MERGE

This is the full triadic dynamics regime.

**Regime III — Non-interacting decay**
(Large R, high tangential velocity, strong dissipation)

- Convergence too slow
- Partition never flattens
- All cores die independently
- No MERGE events

**8.2 Collapse mechanism taxonomy**
Paper 14 identifies six distinct collapse mechanisms:

1. Field-depletion cascade
2. MERGE cascade
3. Drain-and-decay
4. Triadic collision
5. Ghost MERGE
6. Long-lived 2-core orbit

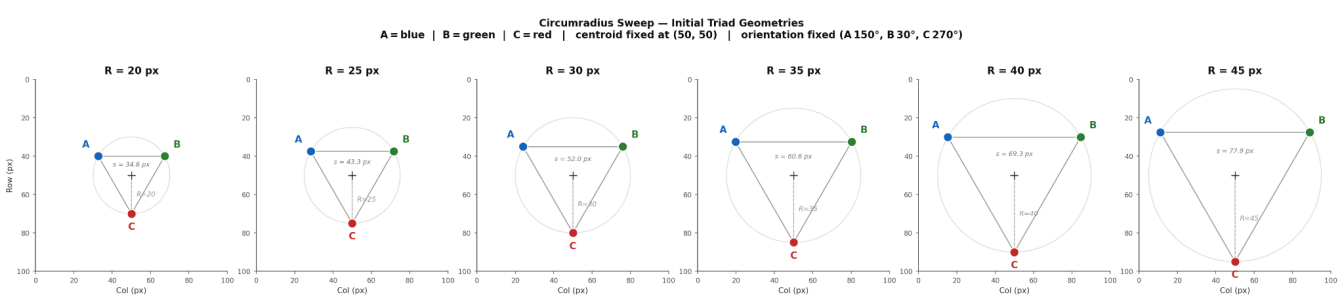
These mechanisms arise from the same underlying geometry:
**the evolution of the three-way partition near the centroid.**

**8.3 Architectural unification**
The six experiments reveal a single architectural truth:

> **Triads collapse because the centroid region cannot support three competing partitions.**
> **Collapse is inevitable; only the mechanism varies.**

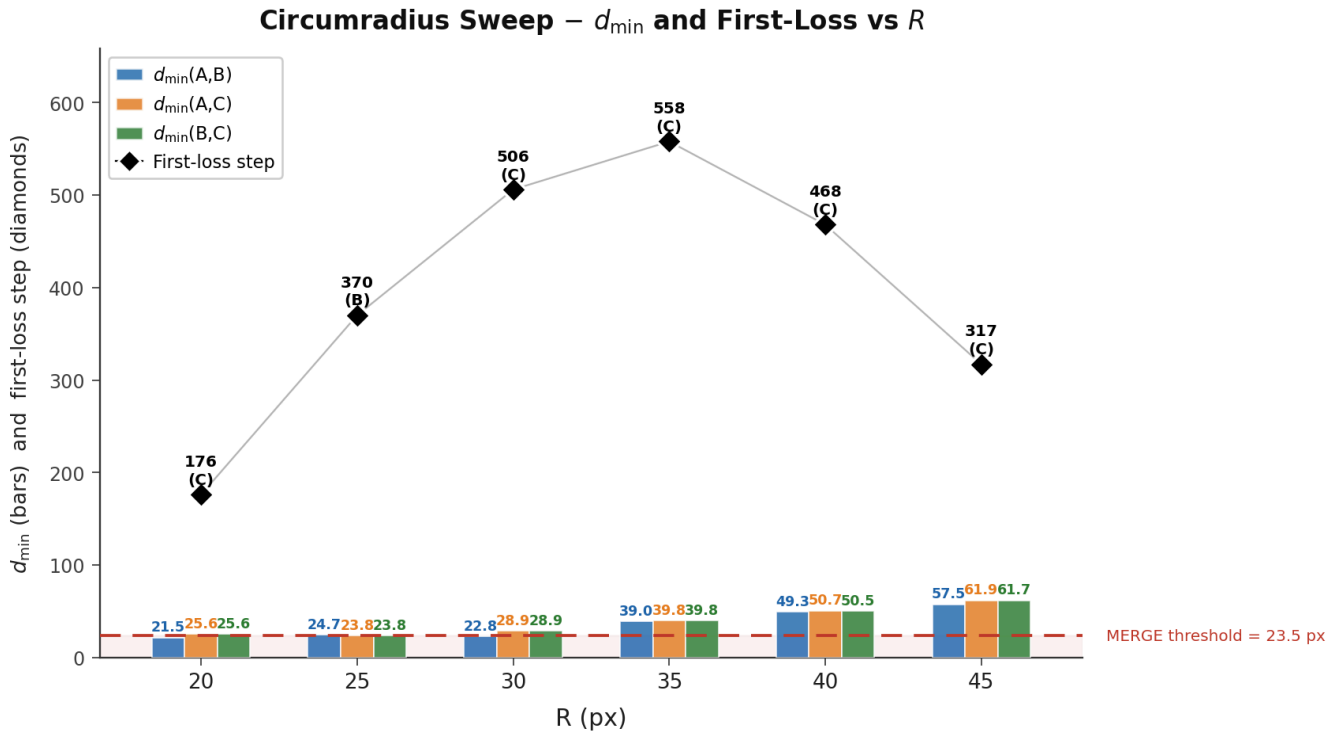
This completes the triadic chapter of ED.

**Figure 11**: R Sweep — Initial Geometries



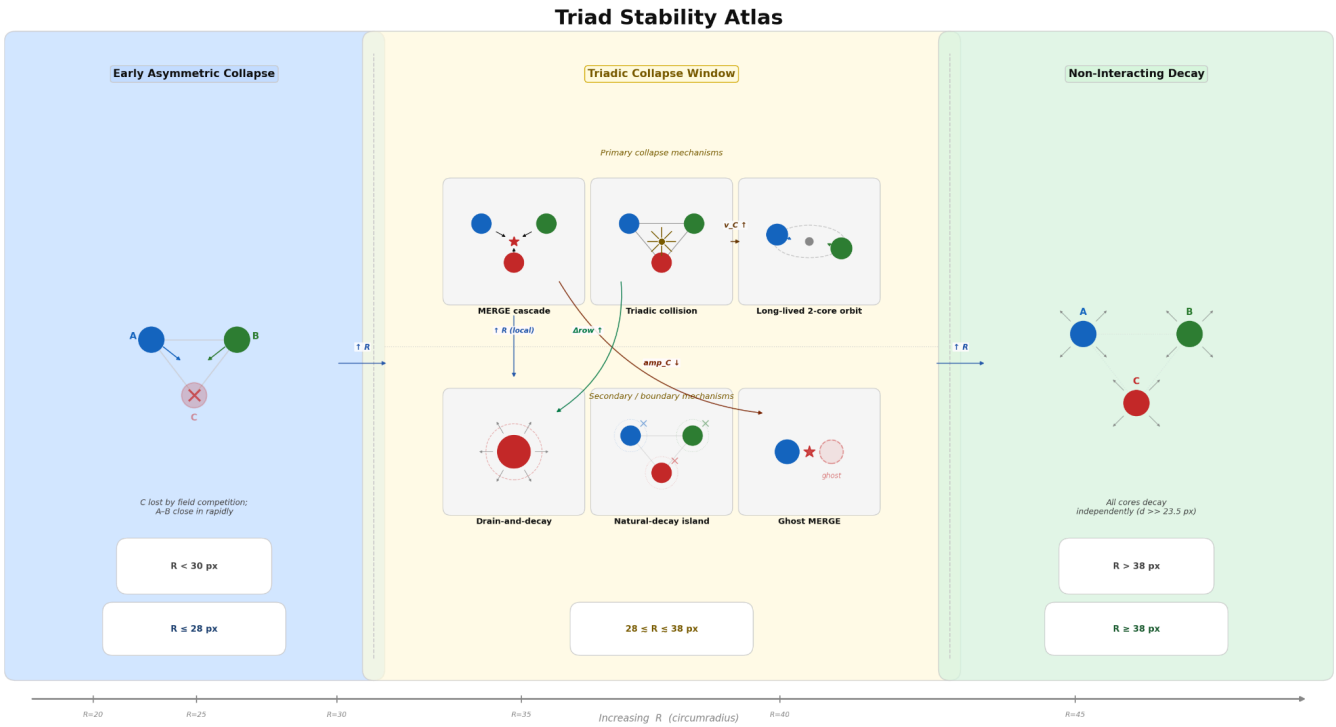
Equilateral triads for R ∈ {20, 25, 30, 35, 40, 45}. Small R produces early asymmetric collapse; intermediate R yields full triadic dynamics; large R produces non-interacting decay.

**Figure 12**: $d_{min}$ and First-Loss vs R




Three-regime structure: long-lived 2-core orbit at R=20; triadic collapse window at R=25–35; non-interacting decay at R≥40.

**Figure 13**: The Triad Stability Atlas




Unified map of all collapse mechanisms across the six experiments. Shows the three master regimes—early asymmetric collapse, triadic collapse window, and non-interacting decay—and the six mechanism types.

## 9. Discussion

### 9.1 Why inward triads fail

Three geometric forces doom inward triads:

1. Centroid flattening
   - The three-way partition becomes planar near the centroid, starving one core.

2. Two-sided squeezing
   - Any core opposite a pair is drained faster than the pair.

3. Convergence–lifetime mismatch
   - If convergence is too slow, α-decay kills the triad before interaction.

These forces appear in every experiment, regardless of α, amplitude, velocity, or R.

### 9.2 Implications for ED architecture

Paper 14 shows that:

- inward triads cannot form stable 3-core molecules
- stability requires breaking inward convergence
- orbital or repulsive regimes are the natural next targets
- triads are the first system where ED's soft-partition geometry becomes fully non-pairwise

This positions triads as the gateway to higher-order ED structures.

9.3 Limitations and future directions
Paper 14 focuses on inward motion. Future work includes:

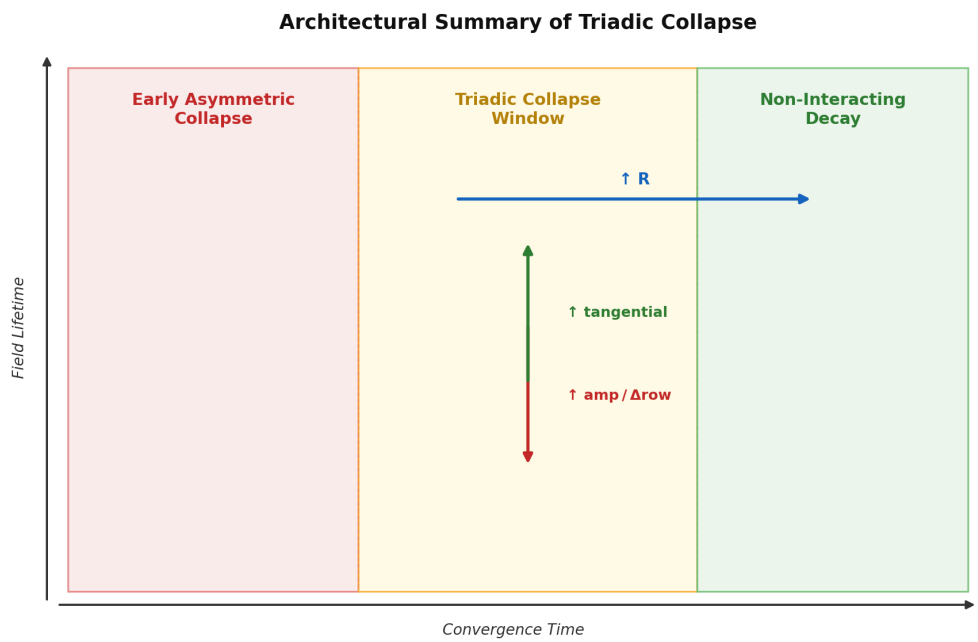
- orbital triads with $v_{tan} > v_{in}$
- repulsive or anti-inward regimes
- 4-core and 5-core stability
- external forcing fields
- adaptive or time-varying amplitudes

These directions extend the architectural program beyond collapse.

## 10. Conclusion

Paper 14 establishes the first complete triad-stability atlas in ED.
Across six experiments spanning dissipation, amplitude, geometry, velocity, and initial separation, we find no stable 3-core inward triad. Instead, the triad occupies a structured set of collapse regimes determined by the geometry of the shared field and the ratio of convergence time to field lifetime.

**Figure 14**: Architectural Summary of Triadic Collapse




Schematic summarizing the architectural laws of ED-Arch-14: collapse is governed by the geometry of the shared field and the ratio of convergence time to field lifetime.

The atlas reveals three master regimes—early asymmetric collapse, the triadic collapse window, and non-interacting decay—and identifies six distinct collapse mechanisms. These results show that triadic stability cannot be inferred from pairwise laws and that inward convergence is fundamentally incompatible with 3-core persistence.

Paper 14 closes the inward‑triad chapter and opens the door to orbital and repulsive architectures, where stable multi‑core molecules may yet emerge.